Multimedia Signal Processing - 2nd module

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1 Question - 10 pts

An outdoor temperature sensor needs to communicate its readings. The admissible temperature range is $x \in [-50, +50]\,^{\circ}\mathrm{C}$.

- Plot the rate-distortion function (SNR-vs-rate) assuming that data is modeled as a continuous source without memory, uniformly distributed in $[-50, 50]$.
- Plot the operational rate-distortion function (SNR-vs-rate) assuming that data is modeled as a continuous source with memory and DPCM is used. Assume that the predictor being used is $\hat{x}_i = x_{i-1}$, In this case, the prediction error e is uniformly distributed in $[-7.5, +7.5]$.

In both cases, find the rate (expressed as an integer number of bits) and the quantization step size Δ when the system is operated at a target $SNR = 30dB$.

Consider the following data sequence

$$\{20, 22, 24, 26, 32, 34, 30\}$$

- Consider a DPCM coding scheme with fixed step size $\Delta = 5$ and the number of levels $M = 3$.
- Consider a backward-adaptive DPCM coding scheme, in which $M = 3$ and

$$\begin{aligned} \Delta &= 3, \quad \text{when } \tilde{e}_{i-1} = 0 \\ \Delta &= 5, \quad \text{otherwise} \end{aligned}$$

In both cases:

- Compute the decoded sequence obtained using such DPCM scheme.
- Compute the number of bits needed to encode the data sequence above when entropy coding is disabled.
- Verify that, for this specific example, the same mean square error distortion is obtained.

Assume that the first symbol is lossless coded and that quantizers have a granular region defined as $[-\frac{M}{2}\Delta, +\frac{M}{2}\Delta]$.

2 Question - 5 pts

Illustrate motion estimation/compensation based on block-matching.

Consider an input $N \times N$ image. Let B denote the block size and W the search range. Express the computational complexity of motion estimation C_{ME} and compensation C_{MC} in terms of number of elementary operations, as a function of N, B and W (with integer pixel accuracy).

3 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Let X, Y denote two sources modeling, respectively, the horizontal and vertical components of the motion vectors in Question 2, read in raster scan.
   - [T] [F] $H(X,Y) < H(X) + (Y)$
   - [T] [F] $H(X) >= \log_2(2W+1)$
   - [T] [F] If motion is mostly due to horizontal panning, $H(Y) < H(X)$
   - [T] [F] X is a memoryless source
2. Consider a perceptual audio coding system
   - [T] [F] The number of bits allocated to each frequency band is inversely proportional to SMR
   - [T] [F] The perceptual model and the analysis filterbank adopt the same time-to-frequency mapping
   - [T] [F] MDCT achieves almost perfect reconstruction
3. Consider a block-based image coding system
   - [T] [F] With zonal quantization, each block is allocated the same bit budget
   - [T] [F] Decreasing the block size, the computational complexity decreases
   - [T] [F] RGB to YC_bC_r mapping addresses inter-pixel correlation.

4 Question - 12 pts

Load data by means of the Matlab command `wavread('stereosynth.wav')`. The file contains notes played on a stereophonic synthesizer.

- Separate the left and right channels and plot them.
- Segment each channel into frames of $T = 0.1$ seconds. For each frame i, estimate the pitch $p_L(i)$ (left channel) and $p_R(i)$ (right channel) with a technique of your choice in the range [30, 300] Hz. Assume integer pitch frequency values.
- Plot the two pitch frequency sequences on a stem plot.
- Compute the conditional entropy $H(L|R)$, where L and R denote, respectively, the left and right channel pitch frequency sequences.